Supplemental notation contract

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Supplemental notation contract

This supplement is the authoritative notation contract for Active Fedference. The methods, formalism, results, figures, report schemas, and API documentation use these meanings even when a source paper uses a different symbol.

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Supplemental notation... (part 2)

A symbol is not reused for a different mathematical object merely because the objects are both probability vectors.

Supplemental notation... (part 3)

The implementation names in the final column are the canonical names for new code and reports; old names survive only as warned, parity-tested compatibility adapters.

Probability objects and generative-model quantities

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States, posteriors, and site factors

Symbol	Meaning
s	Hidden categorical-state index.
o	Observation/outcome index.
$q_n(s)$	Agent n's updated local posterior.
$q(s)$	Server consensus posterior.

States, posteriors, and site... (part 2)

Here $s \in \{1, \ldots, n_s\}$ and $o \in \{1, \ldots, n_o\}$ are indices, not distributions. Each $q_n(s)$ is the local posterior over the shared latent state after agent n's update; $q(s)$ is the corresponding server consensus, and $q_{-n}(s)$ removes agent n's site contribution. The site $t_n(s)$ is a natural-parameter factor.

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States, posteriors, and site... (part 3)

The bridge-only quantity satisfies $m_n(s) = \log q_n(s) + \kappa_n$ for state-constant κ_n; it has no project API and does not imply that every source-protocol message is a broadcast posterior.

States, posteriors, and site... (part 4)

The corresponding code terms are `state`, `observation`, `local_posteriors[n]`, `global_posterior` or `consensus`, `cavity(...)`, and `site_factor`, in table order through $t_n(s)$. The bridge-only $m_n(s)$ has no project API.

Priors, policies, and POMDP quantities

Symbol	Meaning
$\pi_0(s)$	Hidden-state prior, not a policy.

Priors, policies, and... (part 2)

The state index s, posterior $q(s)$, prior $\pi_0(s)$, and policy π must not be conflated. In particular, the policy symbol is bold when needed, never used for a prior, and the prior is never called a policy. The uppercase POMDP tensors A, B, C, D_0 are model objects; they are not posterior factors.

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Priors, policies, and... (part 3)

Each state-indexed column of A is a pmf. The tensor B is indexed by next state, current state, and control. The preferred-outcome pmf is $p_C(o) = \mathrm{softmax}(C)[o]$, and $q(o \mid \pi)$ is the predicted-outcome distribution used in EFE calculations.

Priors, policies, and… (part 4)

The canonical code terms, in table order, are `prior` or `log_prior`, `policy`, `likelihood`, `transition`, `log_preferences`, `initial_prior`, and `predicted_outcomes`.

Priors, policies, and... (part 5)

For the qualified relation to Friston et al.'s Eq. 7 (Friston et al. 2024), the shared support is finite, every $q_n(s)$ is positive on it, the Eq. 7 softmax input is represented by the bridge-only $m_n(s)$ above, and the declared weights w_n are fixed rather than functions of the emerging consensus.

Priors, policies, and... (part 6)

Under exactly those assumptions, additive κ_n constants cancel under softmax and eq. 9 is the categorical posterior-log-potential specialization of the source message-combination term. It neither reconstructs source message construction, cavity/exclusion policy, scheduling,

Priors, policies, and... (part 7)

generative factors, nor the complete source protocol.

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Divergences, losses, and scalar controls

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Generalized-Bayes and aggregation terms

Symbol	Meaning
$\mathcal{D}(q\|p)$	Regularizing divergence.

Generalized-Bayes and... (part 2)

The regularizing divergence acts between distributions and may be KL, reverse KL, or α-Rényi. The positive learning rate/temperature τ multiplies accumulated loss and accepts `learning_rate` only as a warned compatibility alias.

Generalized-Bayes and... (part 3)

The symbols w_n, a_n, and $\tilde{a}_n$ are deliberately distinct. The first is supplied before aggregation, the second is the variational raw server output satisfying $0 \leq a_n \leq w_n$, and the third is only its normalized influence representation returned for interpretation and plotting.

Generalized-Bayes and... (part 4)

Their canonical code terms, in table order, are `divergence`, `loss_by_state`, `tau`, `base_weights[n]`, `raw_effective_weights[n]`, and `normalized_effective_weights[n]`.

Generalized-Bayes and... (part 5)

The server heuristic's reweighting is not a FedGVI client loss and does not inherit a client-side robustness theorem. `robust_aggregate` is a server heuristic with the tested $c = 0$ recovery identity.

Generalized-Bayes and... (part 6)

`variational_aggregate` owns the explicit finite-simplex objective and the raw-weight bound; that bound is not an estimator-level B-robustness theorem.

Robustness, divergences, and loss controls

Symbol	Meaning
$c \geq 0$	Server divergence-reweighting coefficient.

Robustness, divergences... (part 2)

The coefficient λ controls the variational objective and its coordinate updates; $\lambda \downarrow 0$ is a separate deterministic tied-argmax endpoint. The parameter q_{loss} belongs to $L_{q_{\text{loss}}}$;

Robustness, divergences... (part 3)

its $q_{\text{loss}} \downarrow 0$ NLL limit is handled separately, and the subscript prevents collision with posterior $q(s)$. Contamination rate ρ is always interpreted within the declared attack mechanism.

Robustness, divergences... (part 4)

The associated code terms are `robustness`, `entropy_weight`, `alpha`, `beta`, `q_loss`, and `contamination_rate` or `rate`, in table order.

Robustness, divergences... (part 5)

For the objective-backed server rule, the complete scalar-control contract is defined for $c > 0$ and $\lambda > 0$:

Robustness, divergences... (part 6)

$$F_\lambda(q, a) = \sum_n a_n \,\mathrm{CE}(q, q_n) - \lambda H(q) + \frac{1}{c}\,\mathrm{KL}_{\mathrm{gen}}(a \| w). \tag{34}$$

Robustness, divergences... (part 7)

Its two coordinate updates are shown separately so each display remains legible at the presentation typography floor:

Robustness, divergences... (part 8)

$$q \propto \exp\left(\frac{1}{\lambda}\sum_n a_n \log q_n\right), \qquad (35)$$
$$a_n = w_n \exp[-c\,\mathrm{CE}(q, q_n)].$$

Robustness, divergences... (part 9)

The coordinate updates eq. 35 use $\lambda = 1.0$ by default; the `entropy_weight` argument exposes the stated tempered family. The $c = 0$ branch is handled as a recovery limit outside the $c > 0$ objective; at $\lambda = 1.0$ it is the exact project log-linear pool.

Robustness, divergences... (part 10)

The $\lambda \downarrow 0$ endpoint is separately implemented as a deterministic tied-argmax rule and is not obtained by substituting $\lambda = 0$ into the displayed objective or update.

Cavity and factor algebra

For a positive-support global posterior and site term, the cavity operation is defined in log space and then normalized:

Cavity and factor algebra (part 2)

$$q_{-n}(s) = \frac{q(s)/t_n(s)}{\sum_{s'} q(s')/t_n(s')} = \mathrm{softmax}(\log q(s) - \log t_n(s)) \,. \tag{36}$$

Cavity and factor algebra (part 3)

The corresponding factor replacement is

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Cavity and factor algebra (part 4)

$$\begin{aligned} \ell_n(s) &= \log t_n^{\text{old}}(s) \\ &\quad + \log q^{\text{new}}(s) - \log q^{\text{old}}(s), \qquad (37) \\ t_n^{\text{new}}(s) &\leftarrow \operatorname{softmax}(\ell_n)(s). \end{aligned}$$

Cavity and factor algebra (part 5)

Here ℓ_n is the transient, unnormalized log-site vector; the softmax is exactly the exponential normalization of that vector.

Cavity and factor algebra (part 6)

The code-level `cavity` adapter takes `global_posterior` and `site_factor`. The `update_factor` adapter takes `old_site_factor`, `old_global_posterior`, and `new_global_posterior`. The old keywords `posterior`, `factor`, `old_factor`, `old_posterior`,

Cavity and factor algebra (part 7)

and `new_posterior` are accepted only with a `DeprecationWarning`; mixed canonical/old calls fail closed.

Cavity and factor algebra (part 8)

Recombination is tested by normalizing $q_{-n}(s)t_n(s)$ and checking recovery of $q(s)$ to floating-point tolerance. A transported site factor is represented as a pmf, so its arbitrary positive natural-parameter scale is fixed by the explicit normalization above.

Statistical notation and nesting

Symbol	Contractual meaning
n_{seed}	Number of independently seeded worlds/replicates in a declared cell; the inferential unit for seed-level summaries.

Statistical notation and... (part 2)

For the robustness sweep, the primary result is r_{rb} and the matched mean difference $\overline{\Delta}$ with its bootstrap interval. The d-equivalent is retained only as a monotone secondary display and planning input. When $|r_{\mathrm{rb}}| = 1$, the transform diverges;

Statistical notation and… (part 3)

reports use a finite sentinel and captions disclose saturation rather than presenting a million-scale number as a scientifically interpretable effect.

Statistical notation and... (part 4)

Power, prospective sample size, MCSE, and MDE are observed-effect planning/precision diagnostics, not evidence that a confirmatory effect exists.

Statistical notation and... (part 5)

The predeclared headline display rule is the largest positive r_{rb} among robust methods, with the declared method order as a deterministic tie break. A report must also expose the complete tied-method set, the tie-break, the method with the largest mean $\overline{\Delta}$,

Statistical notation and... (part 6)

and the method with the largest mean at the worst rate.

These are distinct summaries; none is a unique scientific winner when the evidence is tied or conditional.

Statistical notation and... (part 7)

For the review grid, every configured robust method remains an inferential member and a displayed rate-profile curve. No pooled-mean selection creates a curve, interval, or hypothesis-test member for that surface.

Code and manuscript naming map

Every arrow names a legacy surface followed by its canonical replacement.

Belief inputs. `beliefs` or `agent_beliefs` → `local_posteriors`; `weights` → `base_weights`. Both are warned keyword/property adapters.

Code and manuscript... (part 2)

Aggregation outputs. Result property `agent_weights` → `normalized_effective_weights`; variational argument `agent_weights` → `raw_effective_weights`. Both are warned adapters, but represent different weight scales.

Code and manuscript... (part 3)

Diagnostics. `shared_beliefs` → `shared_posteriors`; `loss_vec` → `loss_by_state`. These are warned diagnostics and keyword adapters.

Code and manuscript... (part 4)

Statistics. `cohens_d_from_rank_biserial` → `d_equivalent_from_rank_biserial`; report field `cohens_d` → `d_equivalent`. The function is a warned adapter; the report change is a versioned migration.

Code and manuscript... (part 5)

These adapters never silently reinterpret an argument or result. The `agent_weights` result property preserves serialized wire compatibility, while the variational objective and new reports use the canonical weight terms. New reports are canonical and schema-versioned;

Code and manuscript... (part 6)

readers fail closed on unsupported versions. `d_equivalent_from_rank_biserial` returns the declared rank-biserial-derived d-equivalent, not raw Cohen's d; `loss_vec` remains only a warned generalized-Bayes keyword adapter.

Code and manuscript... (part 7)

The wire-level key `agent_weights` is preserved because it is a federation transport contract, not a claim about the scale or meaning of the new result fields. A future wire migration requires an explicit version and a fail-closed reader; it must not silently reinterpret the key.

Source and evidence boundaries

The Friston belief-sharing equations are source equations/protocol claims. Only under the explicit finite-shared-support, posterior-log-potential, and fixed-weight bridge above does the categorical log-linear pool specialize the source message-combination term;

Source and evidence... (part 2)

the tested $c = 0$ identity remains project-local. The generalized-Bayes and loss limits are implementation analogues checked in the finite categorical model.

Source and evidence... (part 3)

The contamination, gallery, onset, conditional-world, and review-grid quantities are conditional simulation evidence over declared cells. None of these finite surfaces is an external-data replication, a reconstructed source protocol, a universal attack taxonomy, a causal intervention,

Source and evidence... (part 4)

or a proof of a server-side robustness guarantee.

Source and evidence... (part 5)

Open theory, calibration, protocol, continuous-state, external-data, authenticated-federation, and clean-release work remains open in the project TODO and claim-audit documents.